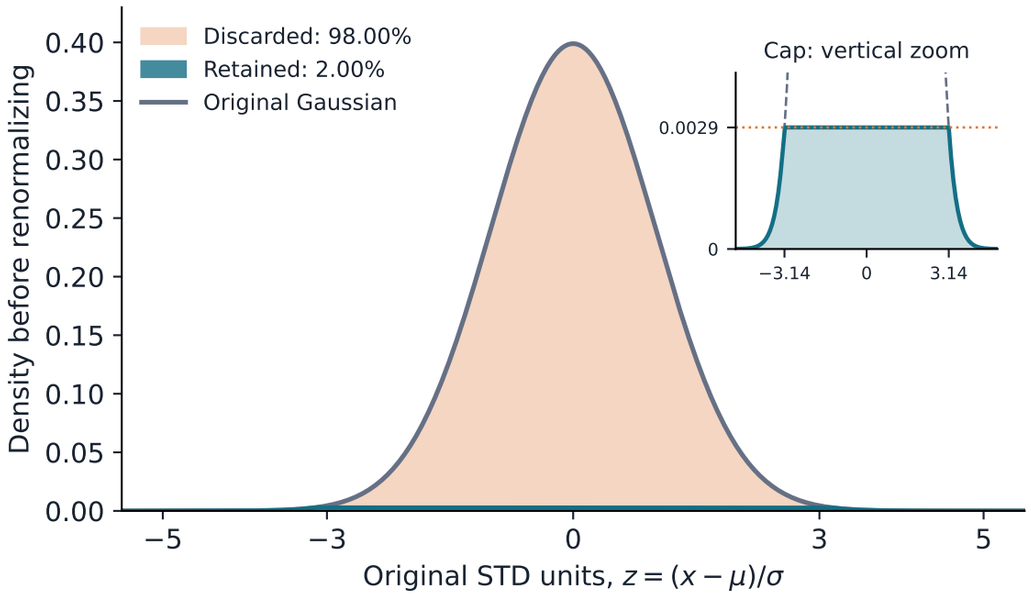
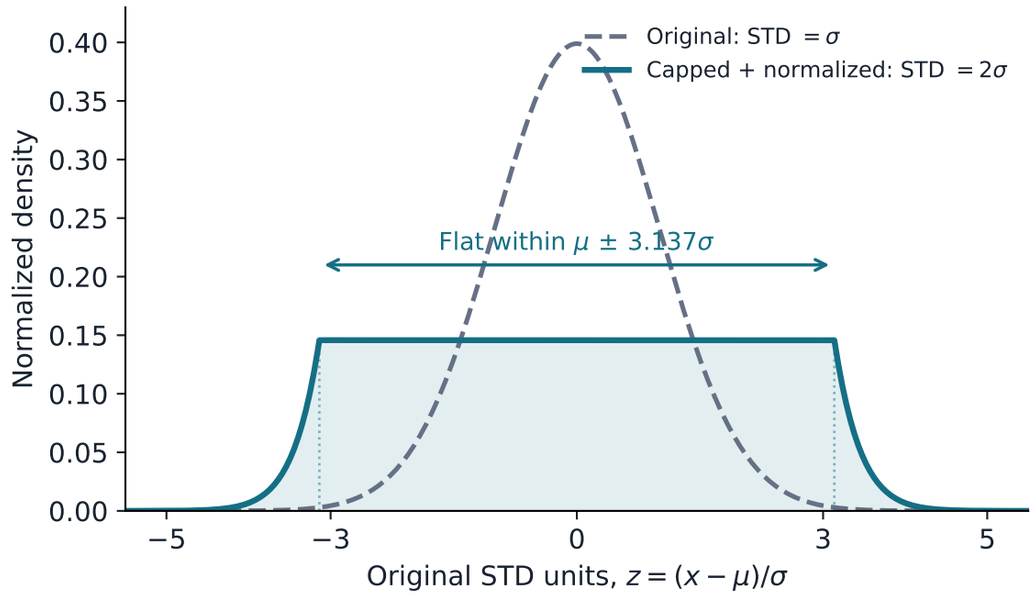
Capping a Gaussian: twice the STD retains only 2.00% of samples

Randomly thin the dense center to a flat cap; keep every tail sample. Symmetry preserves the population mean.

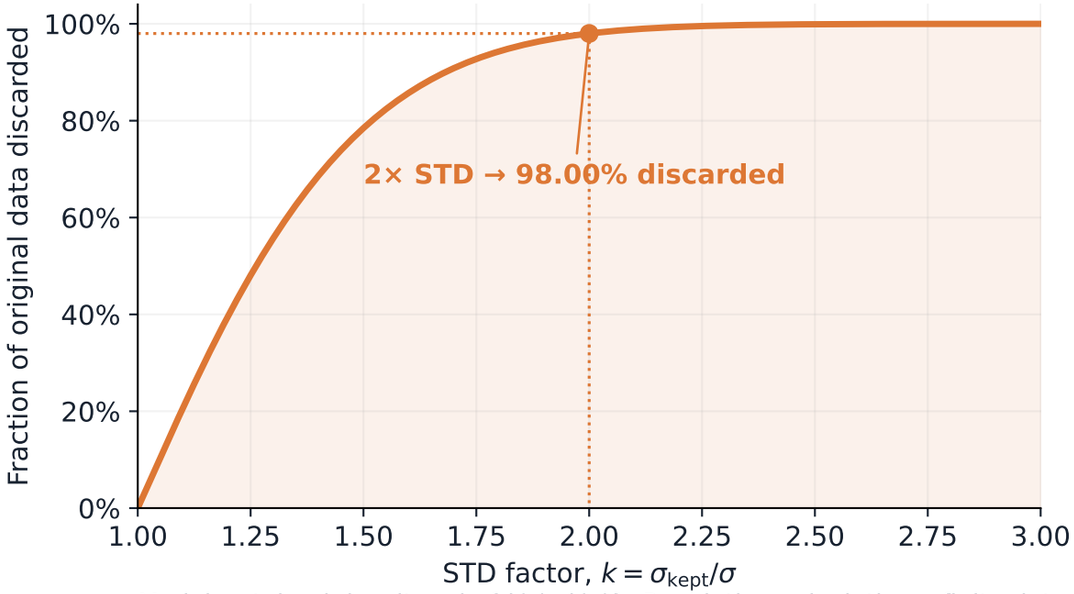
A Remove the peak of the original density



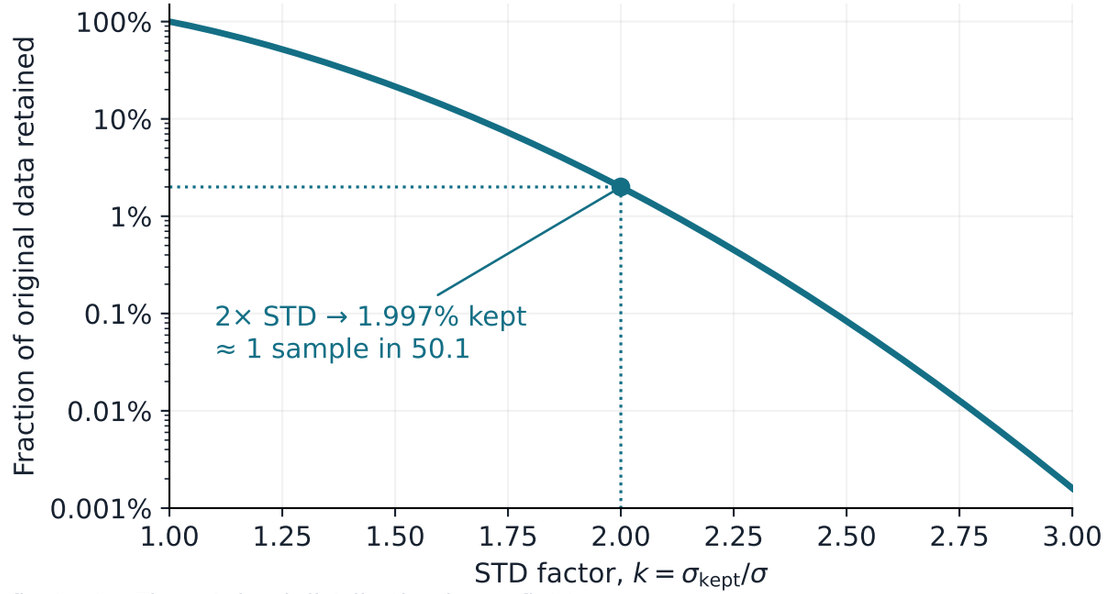
B Renormalize the retained samples



C Data discarded versus target STD factor



D Data retained (log scale shows the cost)



Model: retained density min $\{\phi(z), \phi(a)\}$. Population calculations; finite datasets fluctuate. The retained distribution has a flat top.